



PhyCA-GAN: Physics-causal attention generative framework for imbalanced leak detection in energy transportation pipelines

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ABSTRACT

The scarcity of leakage samples and the resulting class imbalance represent a major challenge for data-driven energy transportation pipeline leak detection methods. To overcome this problem, this paper proposes the physics-causal attention generative adversarial network (PhyCA-GAN), a novel framework that embeds spatio-temporal causality into generative models to generate leakage samples that are both statistically faithful and physically plausible. Specifically, the framework introduces a kinematic-prior encoder (KPE) to fuse physical priors with data features via a learnable knowledge base, thereby establishing strict kinematic boundaries for the generation process. In order to capture complex signal dynamics, a physically-decomposed causal generator (PDCG) is designed to separate leakage signals into macro-physical trends and micro-turbulence details. A novel physics-causal attention mechanism is embedded within this generator, utilizing a dynamic kinematic mask to enforce Saint-Venant propagation constraints and eliminate non-physical spatiotemporal artifacts. Moreover, a physically invariant discriminator (PID) splits the evaluation procedure to guarantee the compliance with the laws of statistical fidelity and physical conservation. The experimental findings suggest that PhyCA-GAN produces better synthetic data than the state-of-the-art baselines and augmented datasets obtained with it enhance the strength and accuracy of downstream leak detectors significantly.

1. Introduction

Cross-regional oil and gas transport is based on energy transportation pipelines, which makes leak detection an important issue in energy security [1] [2] [3]. Although deep learning has redefined this area, its implementation in the industrial environment has two significant bottlenecks. To begin with, the intensive use of large labeled datasets is incompatible with the real-world scenario of sparse and imbalanced leakage information [4] [5]. Second, black box algorithms are not as opaque as required by strict engineering standards of system reliability and interpretability [6] [7]. Hence, developing models that are both accurate and trustworthy—even with imbalanced data—is a key challenge in moving leak detection from simple "perception" to "cognition".

Strategies for solving data imbalance in pipeline leak detection can be roughly divided into sampling-based methods and generative models [8] [9] [10]. Although sampling methods, including undersampling and

oversampling, are the dominant factors in industrial applications, they have inherent deficiencies in terms of handling time series data. These methods usually implement data-level rebalancing but usually do not consider spatiotemporal correlations that are inherent in pipeline signals [11] [12]; In contrast, generative models are a more promising approach. By knowing the underlying probability distribution of the data, they hope to create good quality synthetic samples to effectively solve the problem of data imbalance.

With the fast development of AI, generative models have become an effective alternative to traditional sampling methods for pipeline leak detection. The mainstream models mainly contain variational autoencoders (VAEs), diffusion models and GANs [13] [14]. Among them, VAEs have been widely explored for their capacity of learning compact latent representations [15] [16]. For example, Zhang et al. [17] used VAE in conjunction with transfer learning to create a multi-task model that successfully performed leak detection and localization in few-shot

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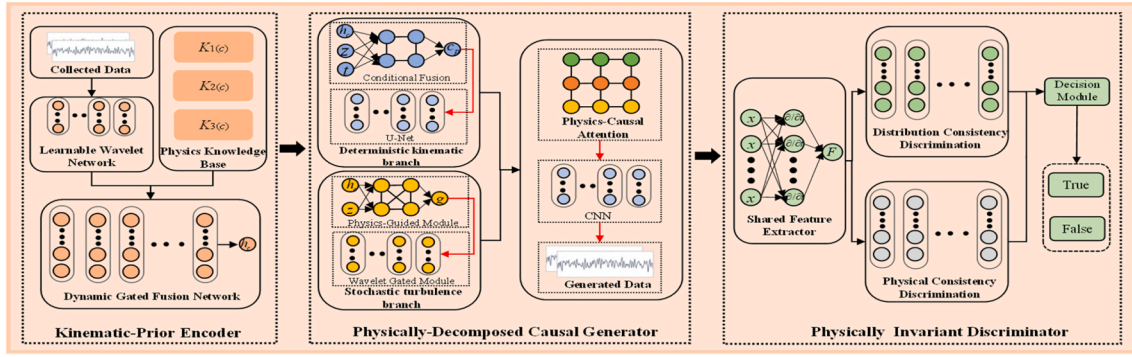


Fig. 1. Schematic diagram of the PhyCA-GAN.

conditions. Similarly, Ref. [18] combined adversarial autoencoders with generative network, which successfully addressed the data imbalance and noise problems in the detection of leaks by the improvement of feature representation; However, despite the stability of training and the ease of probabilistic interpretations, VAEs and their variants tend to produce samples with a lack of detail fidelity and the risk of posterior collapse. This bottleneck has led to recent research interest in diffusion models which provide better distribution coverage and generation diversity [19] [20]; In pipeline applications, diffusion-based approaches have shown great efficacy. Yang et al. [21] proposed diffusion-based data augmentation method for magnetic flux leakage (MFL) data. Through using a denoising process to generate synthetic data with defect class information, this method was able to effectively improve the performance of downstream leak detection models; Notably, using physical constraints has also gained traction. Xu et al. [22] used diffusion models together with physics-informed graph convolutional networks (DM-PIGCN) for gas concentration field reconstruction. Ref. [23] developed a novel diffusion model by integrating simulation and experimental data and was able to produce high fidelity signals of certain types of defects; These works together support the potential of diffusion models in addressing the labeled data scarcity in industrial scenarios.

Although diffusion models are good at generating quality, the high computational cost and inference latency makes it difficult to satisfy the real-time requirements of industrial online diagnosis; In contrast, GANs have a unique advantage in resource-constrained edge computing and real-time systems due to the extreme efficiency of their single-step forward generation [24] [25]. Yet, GANs suffer from training instability as well as mode collapse by nature. To overcome these problems, different improved algorithms have been suggested by researchers. For instance, Zhang et al. [26] proposed a dual-frequency cross-feature diffusion GAN. This model can be used to generate a variety of automatically labeled images from a single real image and uses frequency decomposition with special modules to improve the image generation quality to effectively improve the accuracy and reliability of pipeline detection. Wang et al. [27] proposed the subdomain-alignment adversarial self-attention network, which uses temporal correlation and prototype alignment to

optimize the long-range dependency modeling. Similarly, Ref. [28] proposed a data augmentation method named Res-CosGAN for the generation of pipeline leak data. Its main innovation is a generator which combines residual modules with cosine similarity loss, which greatly improves the quality of the synthetic data and the efficiency of training. Unfortunately, while these methods emphasize a great deal on enhancing statistical fidelity and distribution diversity, they ignore the critical physical consistency constraints inherent in energy pipeline systems. For the specific task of leak detection in pipelines, it requires high-quality generated data to meet a set of tripartite coupling constraints: data fidelity, distribution consistency, and physical consistency. Generated data without constraints caused by physical mechanisms will inevitably not be able to break through the bottlenecks of generalization performance and accuracy for diagnostic models.

This paper proposes the PhyCA-GAN in order to overcome the above-mentioned limitations and to address the trade-off between statistical fidelity and physical validity. The major contributions of this paper are summarized as follows:

- (1) We present the PhyCA-GAN, a new framework that creates a causality-aware data generation paradigm for pipeline leak detection.
- (2) We design a KPE that goes beyond traditional hard-coded constraints. It extracts learnable kinematic priors to act as the inductive bias for the generation process.
- (3) We create a PDCG architecture building upon one physics-causal attention mechanism; This innovation is a mathematically enforceable signal propagation limit, and is also physically valid and reduces generative problems such as pattern collapse.
- (4) We propose a PID to realize a novel mechanism for independent evaluation of physical consistency and statistical realism, which is based on a dual-branch architecture and multi-scale feature extraction mechanism.

2. Proposed method

Pipeline leakage sample generation is faced with a major challenge

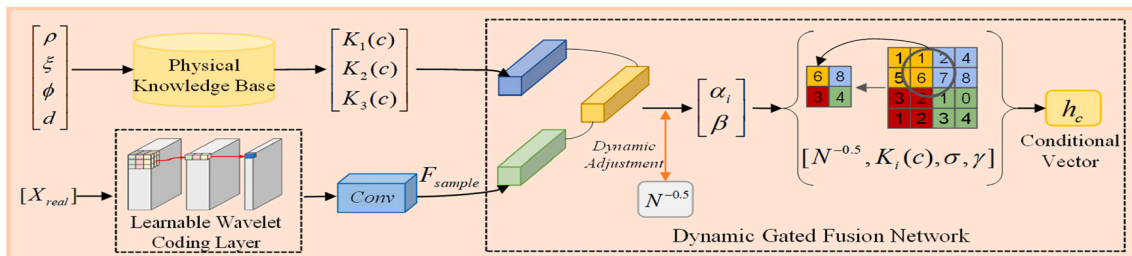


Fig. 2. Schematic diagram of the KPE.

Table 1
Physical knowledge base.

Knowledge Type	Symbol	Physical Interpretation	Selection Basis
Wave velocity characteristics	$K_1(c)$	Pressure wave propagation velocity	Over 90% of the variance in leakage characteristics is due to wave speed variation.
Attenuation characteristics	$K_2(c)$	Energy decay rate	Key metrics for distinguishing leaks from normal conditions
Reflective characteristics	$K_3(c)$	Reflection characteristics of leak points	Core parameter for leak localization

in this regard: current models do not always ensure statistical authenticity and physical consistency. To address this limitation, here we propose the PhyCA-GAN, which is a novel framework that is designed to synthesize high-fidelity leakage samples that are both statistically authentic and physically consistent. This is achieved by co-optimizing three fundamental parts: the KPE module, the PDCG module, and the PID module. The proposed architecture is shown in Fig. 1.

2.1. Kinematic-prior encoder

The KPE addresses pattern collapse and reduces physical distortion by incorporating physical knowledge as the learnable adaptive conditions in the generation process. At its heart, a dynamic gated fusion network adjusts the interaction between physical priors and data features. This mechanism enables a smooth switching between physics-informed and data-driven generation modes, and therefore improves the robustness of the models. Structurally, the KPE is composed of a learnable wavelet coding layer, a physical knowledge base, and the above-mentioned fusion network. The structure of the KPE is shown in Fig. 2.

(1) Learnable wavelet coding layer

The KPE combines a learnable wavelet coding layer for multi-scale feature extraction. Crucially, this module makes the coefficients of the wavelet filters a trainable variable, which makes them easier to adaptively optimize during end-to-end training. This inherent capability fundamentally eliminates the laborious manual wavelet basis selection inherent in the conventional approaches, and enables the network to automatically capture task-specific optimal features.

First, the coefficients of the learnable wavelet filter are initialized

according to db4 wavelet basis. A three-level wavelet packet decomposition is then performed for the input raw signal $X_{real} \in \mathbb{R}^{B \times 1 \times L}$ to extract relevant time-frequency domain features for each level.

$$\begin{aligned} \text{Level1} : A_1, D_1 &= \text{DWT}(X_{real}) \\ \text{Level2} : A_2, D_2 &= \text{DWT}(A_1) \\ \text{Level3} : A_3, D_3 &= \text{DWT}(A_2) \end{aligned} \quad (1)$$

where, A_i and D_i represent the approximation and detail coefficients, which capture the low-frequency overall trend as well as the high-frequency transient characteristics of the pressure wave.

The decomposition results from all three levels are concatenated, followed by compression via a convolutional layer as well as global average pooling (GAP), thus yielding the sample feature f_{sample} .

$$\begin{aligned} F_{concat} &= \text{Concat}[D_1, D_2, D_3, A_3,] \\ F_{sample} &= \text{GAP}(\text{Conv}_{1 \times 1}(F_{concat})) \end{aligned} \quad (2)$$

To make sure the mathematical completeness of the wavelet decomposition as well as to avoid information loss, an orthogonal loss is incorporated into the objective function.

$$L_{ortho} = \|W_a W_a^T + W_d W_d^T - I\|_F^2 + \|W_a W_d^T\|_F^2 \quad (3)$$

where, W_a and W_d represent the approximation and detail coefficients of the wavelet basis matrix. I denote the identity matrix.

(2) Physical knowledge base

Using the actual pipeline parameter distribution, we built a physical knowledge base with the three most leakage-sensitive quantities as shown in Table 1; Their encoding as trainable vectors in a kinematic-prior encoder enabled data-driven prior tuning, thus enhancing model generalization as well as practical utility.

(3) Dynamic gated fusion network

The dynamic gated fusion networks use a gating mechanism to quantify the relative importance of data-driven features as well as physical knowledge. Furthermore, through introducing prior biases, the networks guide the model to dynamically adjust the fusion ratio for the final conditional vector based on data confidence during training; The corresponding computation is as follows:

$$[\alpha_i, \beta] = \text{soft max} \left(W_g^T \cdot [K_1(c)^T; K_2(c)^T; K_3(c)^T; F_{sample}; N^{-0.5}] \right) \quad (4)$$

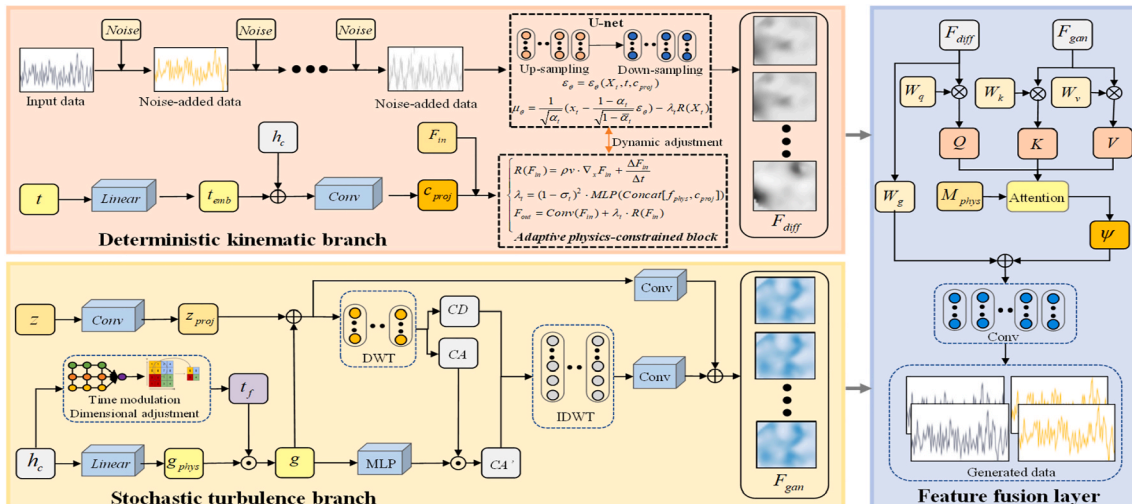


Fig. 3. The structure of the PDCG.

where, α_i and β are the weights for physical knowledge and data features, respectively, where $i = 1, 2, 3$. W_g denotes the weights of the gating network. c indicates the fault type. $N^{-0.5}$ is the reciprocal of the square root of the sample size, used herein as a prior bias to adjust the sample contribution. This design makes sure that a smaller sample size results in a larger $N^{-0.5}$, thus building a reciprocal relationship with the physical knowledge weight α_i .

Through dynamically fusing physical and data features via the prior bias $N^{-0.5}$, the dynamic gated fusion network effectively suppresses overfitting caused by data noise in imbalanced scenarios, thus enhancing robustness while also making sure interpretability via physically meaningful feature representations.

The gating weight amalgamates data features and physical knowledge into a unified conditional vector h_c , which then functions as the governing input to the generator, thus making sure the synthesis process remains constrained by both data-driven patterns and physical laws.

$$h_c = \sum_{i=1}^3 \alpha_i K_i(c) + \beta \cdot \sigma(\gamma N^{-0.5}) \cdot F_{sample} \quad (5)$$

where, γ is the learnable adaptation rate parameter, and σ is the Sigmoid function.

In addition to the fused vector h_c , the KPE extracts the wave velocity parameter $\hat{v}$ (from $K_1(c)$), which is used to control the dynamic kinematic mask.

2.2. Physically-decomposed causal generator

The generator of PhyCA-GAN is a dual branch generator with a deterministic kinematic branch that captures the global wave propagation trends governed by the Saint-Venant equations, and a stochastic turbulence branch that reconstructs high frequency details for higher fidelity. The outputs are then integrated by a physics-causal attention module. The structure of the PDCG is shown in Fig. 3.

(1) Deterministic kinematic branch

1) Condition fusion module

To enable the noise prediction network to be aware of the stages of the generation process, the deterministic kinematic branch encodes the discrete time step t into a continuous vector, providing it with temporal context. This time step vector is subsequently fused with the conditioning vector h_c , and together they are injected as conditions into the noise prediction network, thereby enabling adaptive adjustment of the physical constraint strength and targeted generation of specific fault modes.

$$\begin{aligned} t_{emb} &= \text{Linear}(\sin(W_t \cdot t + b_t)) \\ c_{con} &= \text{Concat}[t_{emb}, h_c] \\ c_{proj} &= W_c c_{con} + b_c \end{aligned} \quad (6)$$

where, W_t and W_c are the frequency matrix and projection matrix, respectively; b_t and b_c denote the phase bias and projection bias vectors.

2) Condition diffusion module

The forward process is formulated as a Markov process governed by the following equation:

$$q(X_t|X_0) = \mathbb{N}(X_t; \sqrt{\bar{\alpha}_t} X_0, (1 - \bar{\alpha}_t)I) \quad (7)$$

where, X_0 is the original input sample, X_t is the noise-added sample at time step t , $\bar{\alpha}_t$ is the cumulative noise retention coefficient, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, α_s is the noise retention coefficient, and I denotes the identity matrix.

Table 2

U-Net network architecture.

Layer	Parameter	Output size
Input		$B \times 1 \times L$
Down sampling 1	Conv1D (64, $k = 7$, $s = 2$) + GN + Swish	$B \times 64 \times L/2$
Down sampling 2	Conv1D (128, $k = 5$, $s = 2$) + GN + Swish	$B \times 128 \times L/4$
Down sampling 3	Conv1D (256, $k = 3$, $s = 2$) + GN + Swish	$B \times 256 \times L/8$
Up-sampling 1	ConvTrans1D (128, $k = 3$, $s = 2$) + GN + Swish	$B \times 128 \times L/4$
Up-sampling 2	ConvTrans1D (64, $k = 5$, $s = 2$) + GN + Swish	$B \times 64 \times L/2$
Up-sampling 3	ConvTrans1D (1, $k = 7$, $s = 2$)	$B \times 1 \times L$

The reverse process is performed by a U-Net, with its architecture detailed in Table 2. To enable precise control over the physical constraints, we introduce an adaptive physics-constrained block $\lambda_t \cdot R(\cdot)$ which provides conditional input to the U-Net. This module dynamically scales the constraint strength with the noise level σ_t : relaxing constraints in high-noise initial phases to encourage diversity, and tightening them in low-noise final phases to guarantee physical consistency. The method makes sure the generated samples are both diverse and physically valid.

The residual term $R(\cdot)$ in our network is mathematically derived from the momentum conservation of the Saint-Venant equations. The standard one-dimensional momentum equation for transient flow in a pipe can be expressed as $\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{f|v|}{2D} = 0$. During the transient propagation of a leakage pressure wave, the convective acceleration of the fluid—represented mathematically by the material derivative $\frac{DF}{Dt} = \frac{\partial F}{\partial t} + v \nabla_x F$ (where F denotes the network's input)—dominates the kinematic evolution. In the U-Net denoising process, to embed this physical restriction into the deep learning architecture, we map the network's latent input to the fluid's kinematic state. By isolating the convective acceleration term and scaling it by the fluid density ρ to maintain momentum dimensions, we abstract the network's internal physical constraint as:

$$R(F_{in}) = \rho v \cdot \nabla_x F_{in} + \frac{\Delta F_{in}}{\Delta t} \quad (8)$$

where, F_{in} denote the input feature maps, $R(F_{in})$ denotes the fluid mechanics residual, ρ is the fluid density, v is the pressure wave velocity, $\nabla_x x_t$ is the spatial gradient, $\frac{\Delta F_{in}}{\Delta t}$ is the time derivative,

In this paper, the time derivative $\frac{\Delta F_{in}}{\Delta t}$ is efficiently discretized via the central finite difference method using a fixed-weight 1D convolutional layer. To calculate the spatial gradient $\nabla_x F_{in}$ without a multi-sensor spatial array, we employ the kinematic relationship of traveling waves. According to the method of characteristics (MOC) in fluid transients, the pressure wave triggered by a leakage propagates bidirectionally along the pipeline. Thus, the spatial and temporal gradients of the propagating feature map are kinematically coupled by the wave velocity v (extracted from the KPE module), formulated as $\nabla_x F_{in} \approx \pm \frac{1}{v} \frac{\partial F_{in}}{\partial t}$. Through this mathematically rigorous substitution, the full spatiotemporal residual is dynamically computed entirely from the 1D temporal tensor.

Ultimately, the output feature map can be represented as:

$$\begin{aligned} F_{out} &= \text{Conv}(F_{in}) + \lambda_t \cdot R(F_{in}) \\ \lambda_t &= (1 - \sigma_t)^2 \cdot \text{MLP}\left(\text{Concat}\left[f_{phys}, c_{proj}\right]\right) \end{aligned} \quad (9)$$

where, σ_t indicates the current noise level, λ_t is the physical constraint weight, f_{phys} is the physical feature sub-vector from conditional vector h_c , and Concat denotes the concatenation operation.

During the reverse pass, U-Net is responsible for predicting the noise added during the forward pass. Since the process cannot be computed directly, a Gaussian distribution is used for approximation:

$$p_\theta(X_{t-1}|X_t) = \mathbb{N}\left(X_{t-1}; \mu_\theta(X_t, t), \sum_t\right) \quad (10)$$

where, $\mu_\theta(X_t, t)$ is the mean function of the reverse process, $\sum_t$ is the covariance matrix, $\sum_t = (1 - \alpha_t) \cdot I$, and X_{t-1} is the denoised sample at time $t-1$.

Yet, the mean predicted by traditional diffusion models cannot make sure that samples strictly adhere to physical laws. To address the limitation, the paper designs a physics-constrained mean based on the aforementioned physics-constrained block, thus directly embedding physical consistency into the core of the generative process.

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \varepsilon_\theta \right) - \lambda_t R(X_t) \quad (11)$$

$$\varepsilon_\theta = \varepsilon_\theta(X_t, t, c_{proj})$$

where, θ represents the U-Net network parameters, and λ_t is the physical constraint weight. During the reverse process, the $\lambda_t R(X_t)$ term corrects the mean μ_θ , thus pulling the sampling direction of X_{t-1} toward regions that satisfy the physical constraints.

(2) Stochastic turbulence branch

The stochastic turbulence branch improves the diversity and high-frequency details of generated samples under physical parameter guidance, while making sure compliance with fluid dynamics laws. Comprising an initial convolution layer, physics-guided module, wavelet gated block, as well as an output convolution layer, it processes a noise vector z and physical condition features h_c .

The initial convolutional layer transforms the input noise z into a high-dimensional, semantically rich latent representation, establishing a diverse basis for subsequent generation.

$$z_{proj} = \text{Conv1D}_{k=7}(z) \quad (12)$$

where, $\text{Conv1D}_{k=7}$ represents a convolution operation with a kernel size of 7.

The physics-guided module incorporates explicit physical parameters to steer the microscopic details of the generated data, aligning them with real leakage characteristics and thereby resolving the disconnect between synthetic details and physical principles.

Extraction of physical parameters: The key physical parameters g_{phys} are extracted from the output h_c of the kinematic-prior encoder:

$$g_{phys} = \text{Linear}(h_c) \quad (13)$$

where, Linear denotes a linear layer.

Frequency prediction: To make sure physical fidelity, this module extracts the dominant frequency range of the leakage signal from the condition vector h_c . As this frequency is the core representation of the leakage physics, it thus directly guides and constrains the spectral characteristics of the generated output.

$$f_{dom} = \text{Sigmoid}(W_{freq2} \cdot \text{ReLU}(W_{freq1} \cdot h_c + b_{freq1}) + b_{freq2}) \times 100 \quad (14)$$

where, W_{freq1} and b_{freq1} represent the weights and biases of the first linear layer, respectively, while W_{freq2} and b_{freq2} denote the weights and biases of the second linear layer. Sigmoid is the activation function, constraining the predicted frequency values within the range [0,1]. The $\times 100$ operation extends the frequency range to [0,100] Hz.

Time modulation and dimensional adjustment: The step synthesizes a physically guided signal g with periodic characteristics through integrating key physical parameters with the temporal factor t_f . The resulting signal g , which combines the interpretability of physical parameters with temporal periodicity, is then used to dynamically modulate the input features of the stochastic turbulence branch.

$$t_f = \sin(2\pi \cdot f_{dom} \cdot t) \quad (15)$$

$$g = W_{final}(g_{phys} \odot t_f)$$

where, $\odot$ is the element-wise product, where physical parameters are multiplied element-wise with the time factor. ξ denotes the learnable scaling parameter, used to balance the contributions of physical parameters and data features. W_{final} is the dimension adjustment matrix, which adjusts the dimensions of the physical guidance vector. The physical guided signal g can enhance leakage features at critical time points.

To achieve detailed alignment between the data distribution and physical laws, the noise feature maps are fused with the physical guided signal g and serve as the input to the wavelet gated block.

$$x_{wav} = z_{proj} + g \quad (16)$$

Subsequently, the input x_{wav} is decomposed using a wavelet transform. Within the transform domain, the approximation coefficients CA are modulated by the signal g , thereby directly infusing physical constraints at the feature level and ensuring the reconstructed details possess the correct physical properties.

$$CA, CD = \text{DWT}(x_{wav}) \quad (17)$$

$$CA' = CA \odot \text{MLP}(g) = CA \odot \text{Sigmoid}(W_m \cdot g + b_m)$$

where, CA and CD denote the approximation coefficient and detail coefficient, respectively, while DWT represent the discrete wavelet transform. W_m and b_m represent the weights and biases.

The output convolutional layer, as the terminal point of the stochastic turbulence branch, integrates and up-samples features from the wavelet-gated blocks to produce the final output, which is rich in high-frequency details.

$$F_{gan} = \text{Conv}(x_{wav}) + \text{Conv}(\text{IDWT}(CA', CD)) \quad (18)$$

(3) Physics-causal attention fusion module

Traditional feature fusion layers often use standard cross-attention, which allows for global feature interactions. However, in pipeline fluid dynamics, pressure wave propagation is strictly constrained by wave velocity. If the attention mechanism allows the current state to attend to future time steps or details beyond this propagation limit, it creates a "spacetime leap" that violates physical causality. This leads to generated samples containing non-physical artifacts.

To address this issue, we transform the simple feature fusion layer into a physics-causal attention fusion module. This module introduces a dynamic kinematic mask. It uses wave velocity parameters from the KPE to embeds fluid propagation rules directly to the attention mechanism.

The theoretical foundation of the dynamic kinematic mask is rooted in the method of characteristics (MOC) for the hyperbolic Saint-Venant equations. In pipeline fluid dynamics, pressure waves propagate at a finite velocity, establishing a strict physical "domain of dependence" constrained numerically by the Courant-Friedrichs-Lewy (CFL) condition. Standard full-attention mechanisms violate this spatiotemporal causality by permitting instantaneous global correlations, which generate non-physical spacetime leaps. The dynamic kinematic mask resolves this by acting as an explicit structural regularizer. By formulating the dynamic bandwidth B using the extracted wave velocity v , the mask M_{phys} restricts the attention matrix strictly within the physical causality cone. This mathematically enforces the CFL condition in the latent space, ensuring the synthesized temporal interactions strictly obey fluid dynamics laws and eliminating non-physical artifacts.

The deterministic branch output F_{diff} acts as the query, and the stochastic branch output F_{gan} as both the key and value. We then generate the feature subspaces through linear projections:

Table 3

Structure of the shared feature extraction layer.

Layer	Parameter	Output size
Conv1D	64, $k = 5$, $s = 2$, $p = 2$, ReLU	$B \times 64 \times L/2$
Learnable Wavelet Conv	64	
Conv1D	128, $k = 3$, $s = 2$, $p = 2$, ReLU	$B \times 128 \times L/4$
Conv1D	256, $k = 3$, $p = 2$, ReLU	$B \times 256 \times L/4$

$$\begin{aligned} Q &= W_q F_{diff} \\ K &= W_k F_{gan} \\ V &= W_v F_{gan} \end{aligned} \quad (19)$$

where, W_q , W_k and W_v are all weight matrices.

By using the wave velocity from the KPE, we establish a dynamic propagation bandwidth B . This defines the maximum temporal scope a pressure wave can reach under specific operating conditions:

$$B = \sigma(\phi \cdot \hat{v} + \beta) \cdot L \quad (20)$$

where, $\hat{v}$ denotes the extracted physical wave velocity parameter, while ϕ and β are both learnable scaling coefficients. σ represents the activation function, and L indicates the signal sequence length.

Based on bandwidth B , we constructed a physical mask matrix $M_{phys} \in \mathbb{R}^{L \times L}$ to mask out correlations that violate physical propagation laws. For any coordinate (i, j) in the attention map, the mask is defined as:

$$M_{phys}(i, j) = \begin{cases} 0, & \text{if } |i - j| \leq B \text{ (Physically Plausible)} \\ -\infty, & \text{if } |i - j| > B \text{ (Causally Violated)} \end{cases} \quad (21)$$

This mask ensures the model strictly follows spatiotemporal causality by zeroing out attention weights outside the valid propagation cone. We then obtain the final fused features using this masked attention mechanism:

$$\psi = \text{Softmax}\left(\frac{QK^T + M_{phys}}{\sqrt{d_k}}\right) \cdot V \quad (22)$$

To ensure the reliability of the generated results, the model employs residual connections to directly inject macro-physical features into the output, thereby providing a strong guarantee of physical consistency throughout the complex feature fusion process. The generated signal can ultimately be expressed as:

$$\begin{aligned} F_{out} &= \text{Conv1D}(\psi + W_g \cdot F_{diff}) \\ X_{gen} &= \text{Conv1D}(F_{out}) \in \mathbb{R}^{B \times 1 \times L} \end{aligned} \quad (23)$$

2.3. Physical invariant discriminator

The discriminator of PhyCA-GAN adopts a dual-branch mechanism which aims to assess physical consistency and statistical authenticity of generated samples at the same time, which solves the drawback of traditional discriminators that only judge "whether a sample is real". It is composed of a common feature extraction layer, a physical consistency discrimination branch and a statistical authenticity discrimination branch.

1) Shared feature extraction layer

The shared feature extraction layer includes convolutional layers and learnable wavelet convolutional layers to extract a shared feature F , from both the generated and the real samples. Its detailed architecture is given in Table 3.

2) Physical consistency discrimination branch

Multiscale gradient feature extraction: In order to address the limitations of single-scale analysis, this module uses multiple parallel groups

of depth-wise separable convolutions with different receptive fields for processing the shared features F . This design allows synchronous recording of high frequency transients, local stationary characteristics and macroscopic propagation patterns. Consequently, it achieves a robust multi scale representation of leakage dynamics, improving detection performance by a great margin.

$$F_{grad}^i = \text{DepthwiseConv1D}(F, k = i) \quad (24)$$

where, *DepthwiseConv1D* denotes a depth-wise separable convolution operation, and k represents the size of the convolution kernel, $i = [3, 5, 7, 9]$.

Learnable leakage response function: This function is built to obtain corresponding attention masks for the multiscale gradient features, which are realized by a group of lightweight adaptive units. This design allows the model to automatically learn different optimal response patterns for different leakage characteristics, and thus generate a set of targeted multi-scale leakage attention masks.

$$M^{(i)} = \text{Sigmoid}\left(\text{Conv1D}\left(\text{ELU}\left(F_{grad}^i\right)\right)\right) \quad (25)$$

where ELU denotes the activation function. Conv1D represents a convolutional layer with a kernel size of 1.

Adaptive feature fusion: Based on a dynamic weighting mechanism, this module adaptively fuses multi-scale leakage features. It assigns context-dependent importance weights to each scale features, thus allowing the model to focus on the most relevant information for a given operational scenario.

$$\begin{aligned} \alpha_i &= \text{Sigmoid}\left(\text{MLP}\left(\text{GAP}\left(\text{Concat}\left[M^{(3)}, M^{(5)}, M^{(7)}, M^{(9)}\right]\right)\right)\right) \\ F_{phys} &= \sum_{i=3,5,7,9} \alpha_i \cdot (F \odot M^{(i)}) \end{aligned} \quad (26)$$

where, *GAP* and *Concat* represent the global average pooling layer and the concatenation operation, respectively.

Physical consistency discrimination: This branch generates the physical consistency score s_{phys} by applying global average pooling and a Sigmoid function.

$$s_{phys} = \text{Sigmoid}\left(\text{Linear}\left(\text{GAP}(F_{phys})\right)\right) \quad (27)$$

where, *Linear* denotes a linear layer.

3) Distribution consistency discrimination branch

This branch integrates multi-scale convolutions, temporal attention, and frequency-domain transformations in a unified framework to evaluate the statistical authenticity of samples.

a) Multi-scale convolution: A parallel convolutional structure with kernel sizes of 3, 5, and 7 is employed to extract disturbance features across different frequency bands. The outputs from these parallel branches are subsequently concatenated.

$$F_{multi} = \text{Concat}(\text{Conv}_{k=3}(F), \text{Conv}_{k=5}(F), \text{Conv}_{k=7}(F)) \quad (28)$$

b) Temporal attention and frequency-domain transformation: Temporal attention is first applied to capture key temporal locations of transient disturbances (e.g., leakage-induced impulses). The enhanced temporal features F_{att} are then converted via a Fast Fourier Transform (FFT) into frequency-domain features F_{freq} , which characterize periodic and other relevant spectral patterns.

$$\begin{aligned} Attn &= \text{soft max}(\text{Conv1D}(F_{multi})) \\ F_{att} &= Attn \odot F_{multi} \\ F_{freq} &= \text{FFT}(F_{att}) \end{aligned} \quad (29)$$

c) Statistical authenticity discrimination: This branch employs a global pooling layer and a fully-connected layer to compute the authenticity score s_{real} .

$$s_{real} = \sigma(\text{Linear}(\text{GAP}(F_{freq}))) \quad (30)$$

2.4. Loss function and training strategy

(1) Generator loss function

The generator's objective is designed to optimize a composite loss function encompassing physical consistency, adversarial realism, diffusion denoising, and feature matching, formulated as follows:

$$L_G = \lambda_1 L_{phys} + \lambda_2 L_{adv} + \lambda_3 L_{diff} + \lambda_4 L_{fm} \quad (31)$$

where, L_{phys} , L_{adv} , L_{diff} , and L_{fm} denote the physical consistency, adversarial, diffusion, and feature matching losses, respectively, while $\lambda_1, \lambda_2, \lambda_3$ and λ_4 are their corresponding weighting coefficients.

Physical consistency loss is used to ensure generated samples adhere to fundamental physical laws:

$$L_{phys} = \mathbb{E}_{X_{gen} \sim P_g} [\|R(X_{gen})\|_2^2] \quad (32)$$

where, $R(\cdot)$ represents the physical residual.

The adversarial loss is employed to optimize the statistical distribution of the generated samples through the discriminator's authenticity branch, with the objective of maximizing the probability that they are judged as authentic.

$$L_{adv} = -\mathbb{E}_{X_{gen} \sim P_g} [D_{real}(X_{gen})] \quad (33)$$

The objective of diffusion loss is to optimize the denoising capability of the diffusion branch, consistent with conventional denoising diffusion probabilistic models (DDPM):

$$L_{diff} = \mathbb{E}_{t, x_0, \varepsilon} \left[\left\| \varepsilon - \varepsilon_0 \left(\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon, t \right) \right\|^2 \right] \quad (34)$$

Feature matching loss serves to measure the discrepancy between feature representations. This loss function shifts the focus from direct raw data comparison to matching their statistical or structural properties.

$$L_{fm} = \mathbb{E} [\|\phi(X_{real}) - \phi(X_{gen})\|_1] \quad (35)$$

where, $\phi(X_{real})$ and $\phi(X_{gen})$ represent the expected values of the L2 norm between corresponding feature vectors extracted from real and generated samples.

To address training instability caused by inconsistent objectives between diffusion loss and adversarial loss in generators, this paper proposes a dynamic gradient coordination mechanism. This mechanism dynamically adjusts the contribution weight λ_2 of the adversarial loss by calculating the cosine similarity between the gradients of the two losses.

$$\lambda_2 = \begin{cases} 0.1 & \cos \theta < -0.3 \\ 0.5 - 0.3 \leq \cos \theta \leq 0.3 \\ 1.0 & \cos \theta \geq 0.3 \end{cases} \quad (36)$$

$$\cos \theta = \frac{\langle \nabla L_{diff}, \nabla L_{adv} \rangle}{\|\nabla L_{diff}\| \cdot \|\nabla L_{adv}\|} \quad (37)$$

where, ∇L_{diff} is the diffusion gradient, and ∇L_{adv} is the adversarial gradient.

(2) Discriminator loss function

The discriminator's loss function encompasses the dual objectives of physical consistency assessment and statistical authenticity discrimination. The combined objective takes the following form:

$$L_D = L_{adv} + \lambda_{phys} L_{phys} + \lambda_{gp} L_{gp} \quad (38)$$

where, L_{gp} denotes the gradient penalty term, while λ_{phys} and λ_{gp} are the loss weighting coefficients.

The adversarial loss is employed to enhance training stability by utilizing the Wasserstein distance to measure the divergence between the distributions of real and generated samples:

$$L_{adv} = \mathbb{E}_{X_{gen} \sim P_g} [D_{real}(X_{gen})] - \mathbb{E}_{X_{real} \sim P_r} [D_{real}(X_{real})] \quad (39)$$

The physical consistency loss ensures that the discriminator's physics branch accurately scores both real and generated samples, thereby strengthening its discriminative capability in assessing adherence to physical laws.

$$L_{phys} = \mathbb{E}_{X_{real} \sim P_r} [\|D_{phys}(X_{real}) - 1\|_2^2] + \mathbb{E}_{X_{gen} \sim P_g} [\|D_{phys}(X_{gen}) - 0\|_2^2] \quad (40)$$

The gradient penalty term is applied to enforce the Lipschitz constraint, thereby stabilizing the training process and mitigating overfitting in the discriminator.

$$L_{gp} = \mathbb{E}_{\hat{X} \sim P_{\hat{X}}} [(\|\nabla_{\hat{X}} D(\hat{X})\|_2 - 1)^2] \quad (41)$$

where, $\hat{X}$ represents the interpolation of the sample.

(3) Model training strategy

To ensure the stable convergence of the PhyCA-GAN model, in this paper, a phased collaborative training strategy is adopted, which is proposed as follows.

1) Pre-training

This stage aims to provide effective parameter initialization for subsequent joint adversarial training and to mitigate instability during the initial training phase. The process consists of the following two steps:

- KPE pre-training: Freeze generator and discriminator; optimize only KPE. This stage is there to instill preliminary physical interpretability into the learnable wavelet and create an initial representation for the basic knowledge base.
- Deterministic kinematic branch pre-training: Freeze stochastic turbulence branch, KPE and discriminator; train the deterministic kinematic branch only. The goal is to create coarse samples that are physically plausible and structurally coherent and provide a basis for the subsequent adversarial training.

2) Joint adversarial training

In this phase, all the modules are jointly optimized in an end-to-end fashion using the composite loss function. A major challenge is the gradient conflict between the diffusion and adversarial losses of the generator. To overcome this, we use a dynamic gradient coordination mechanism that dynamically balances the contributions of these competing goals, thereby ensuring the coordinated convergence.

3. Experiments

3.1. Experiment setup

The pipeline leakage data set used in this study was acquired from a dedicated experimental platform to validate the proposed model (PhyCA-GAN). The platform consists of an industrial host computer, pipelines, storage tanks and auxiliary components. The total length of the pipeline is 2800 m, and operating pressures are kept at 1.8 MPa at the upstream end and 0.6 MPa at the downstream end. Acoustic pressure sensors are installed at both ends of the pipeline to record acoustic

Table 4
Classification label.

Fault type	Label name
Normal	0
Large Leakage	1
Medium leakage	2
Small leakage	3

signals in different operation conditions, such as large, medium, and small leakage operation, and normal operation. All signals were sampled at a frequency of 1000 Hz. Data acquisition was accomplished with an NI DAQ 9181 system which synchronously sent the acquired signals to the industrial host computer for analysis and storage. Table 4 offers the specific classification labels for each of the pipeline operating conditions, and a detailed illustration of the experimental platform is provided in Fig. 4.

Prior to model training, the raw data was preprocessed. A zero-phase high-pass filter was applied to remove sensor baseline drift. The continuous acoustic signals were then divided into discrete segments of length $L = 1024$ using a sliding window with a 50% overlap. This specific length was chosen because it captures the periodic variations of the pressure wave while matching the model's input dimensions. Following segmentation, Z-score normalization was applied to each sample.

The key hyperparameters of PhyCA-GAN were optimized through a hierarchical grid search combined with cross-validation. The diffusion loss was set as the baseline objective ($\lambda_3 = 1$). To prevent premature convergence to physically invalid solutions, the physical consistency weight of the generator was linearly annealed from 0.1 to 1.0 over the first 30% of training. The adversarial weight λ_2 was dynamically adjusted using the proposed gradient coordination mechanism (Eq. (36)). To ensure reliable physical consistency evaluation, λ_{phys} was fixed at 1.0, while the gradient penalty coefficient λ_{gp} was set to 10.0. The model was trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 64. Table 5 summarizes these settings.

To evaluate the effectiveness of the proposed PhyCA-GAN algorithm, it was benchmarked against six state-of-the-art models. The baseline models were adapted to fit the specific requirements of the pipeline diagnostic task.

GAN [5]: Uses a minimax game between a generator and a discriminator, in order to minimize the Jensen-Shannon (JS) divergence between the distributions of the real and generated data.

WGAN-GP [29]: Uses the Wasserstein distance instead of JS divergence and adds a gradient penalty term that imposes Lipschitz continuity on the discriminator to improve training stability and sample quality.

Phy-GAN: Incorporates a physical loss term into the standard GAN loss function, but does not employ any other complex design.

GPSC-GAN [30]: A data augmentation model for multi-class fault situations, which produces high-quality multimodal samples by combining independent classifiers and a gradient penalty mechanism.

IVQ-LDM [31]: Supports efficient and high-quality fault data

generation by first compressing input data into a latent space, then taking a diffusion model through interpretable kernel-based conditioning.

CSGAN-GP [32]: A conditional adversarial network with semi-supervised learning and gradient penalty using a dynamic threshold filtering strategy to ensure stable generation of high-quality fault samples.

Diff-MTS [33] Generates high-fidelity industrial multivariate time-series data with complex temporal dependencies using adaptive conditional control and temporal feature decomposition.

PI-CGAN [34]: A physics-informed conditional generative adversarial network that enforces dual statistical and physical consistency by embedding governing device equations as implicit generator constraints.

PCHT-GAN [35]: A physics-coupled hybrid transformer generative adversarial network that produces physically consistent leakage data via a coordinated mechanism-prediction and data-compensation paradigm, integrated with dynamic low-rank bilinear spatiotemporal attention.

All the experiments were implemented using the PyTorch package and the following hardware: Nvidia RTX 4060 GPU and Intel(R) Silver 4210 CPU. The model hyperparameters were optimized and determined using cross validation.

3.2. Qualitative evaluation

3.2.1. Time domain feature fidelity analysis

We compare the time-domain waveforms generated by each baseline model against our proposed method, as shown in Fig. 5. The analysis indicates: 1) Standard GAN and WGAN-GP show noticeable waveform distortions compared to real data. This is largely attributed to the inherent instability of GAN training, while WGAN-GP is limited by the sensitivity of its gradient penalty mechanism to hyperparameters; 2) While improved variants based on GAN (CPSC-GAN and CSGAN-GP) alleviate these distortions, they remain susceptible to noise and interference. Without a good domain knowledge guidance, these models tend to generate data with varying quality; 3) However, diffusion-based models (IVQ-LDM and Diff-MTS) have better generation quality. In specific, Diff-MTS is outperformed only by PI-CGAN, PCHT-GAN, and PhyCA-GAN. Nevertheless, it continues to face difficulties with fine-grained feature learning, which causes the loss of important details and small differences in the waveforms; 4) While Phy-GAN improves the quality of generated data by introducing a physical loss term into the loss function, it still suffers from modal overlap, resulting in poor data quality; 5) Benefiting from the incorporation of physical information,

Table 5
Key hyperparameter settings.

Hyperparameter	λ_1	λ_2	λ_3	λ_4	λ_{phys}	λ_{gp}
Value	0.1-1.0	Dynamic (0.1/0.5/1.0)	1.0	0.1	1.0	10.0

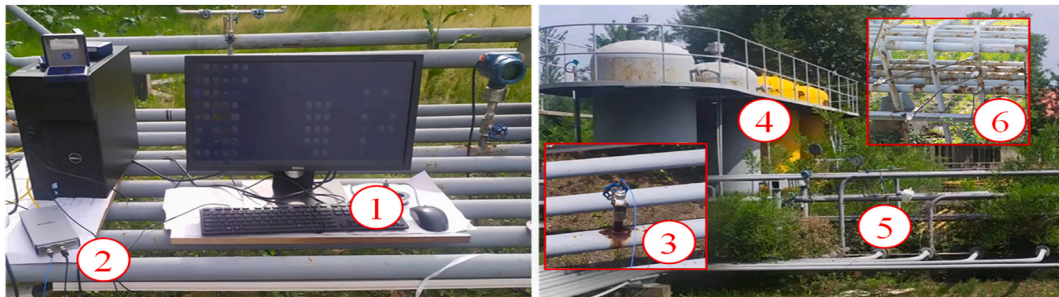


Fig. 4. Schematic diagram of the experimental platform.

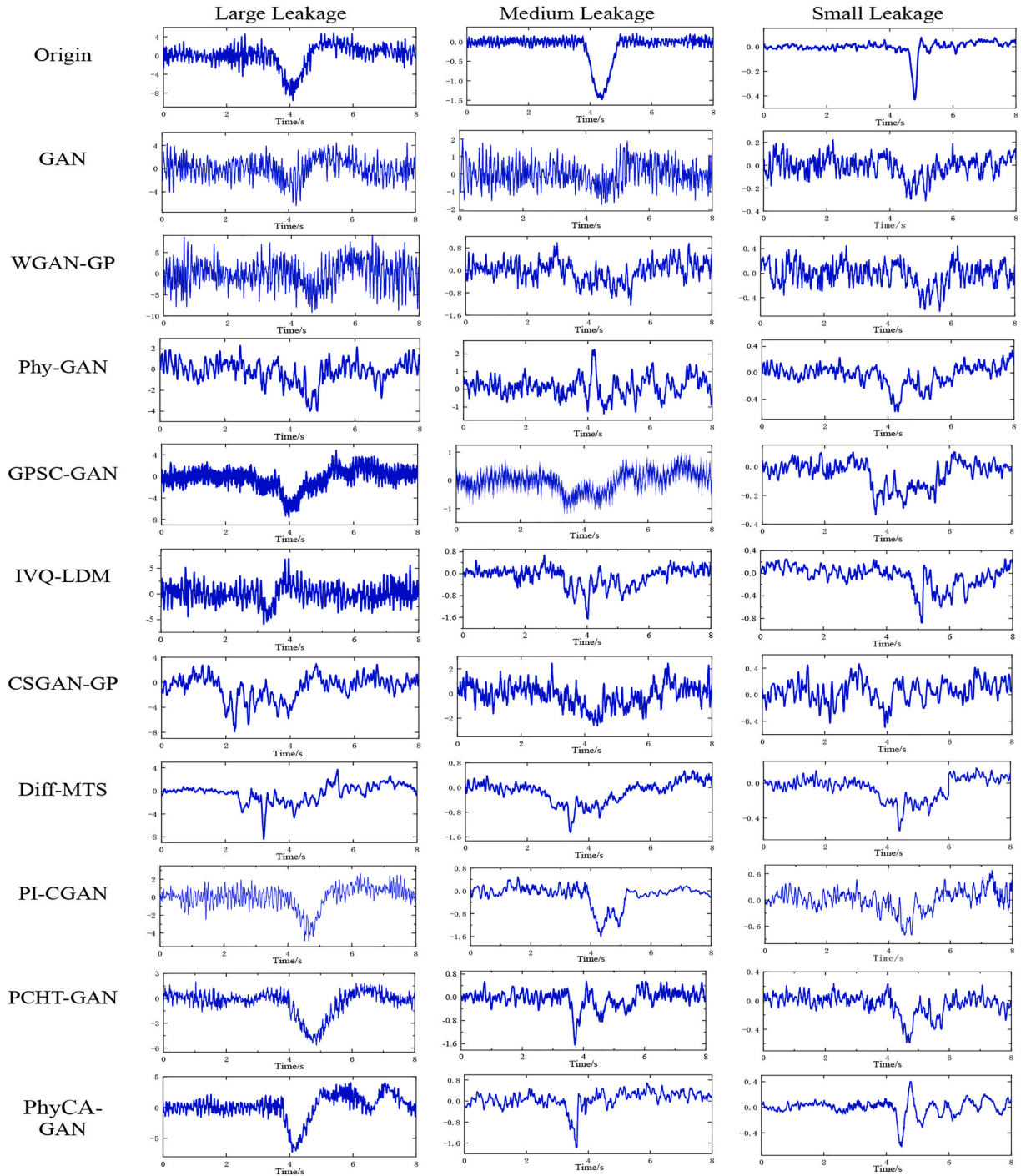


Fig. 5. Time-domain curve.

both the PI-CGAN and PCHT-GAN models exhibit significantly improved performance compared to the preceding models. Nevertheless, constrained solely by the PDE residual, these two models are prone to generating non-physical 'non-causal artifacts', which compromise data fidelity; 6) PhyCA-GAN is highly effective in overcoming pattern collapse and physical distortion. It adapts the physical priors with data features using the KPE. More so, the PDCG records both the coarse and fine details, which greatly enhance the quality of the data.

3.2.2. Distribution and physical consistency analysis

In order to evaluate the data generated by the PhyCA-GAN model quantitatively, we consider four metrics, which are maximum mean

discrepancy (MMD), Kullback-Leibler divergence (KLD), cosine similarity (Cosine) and physical residual (Res). MMD and KLD are used to measure the difference between the generated and real data of the statistical distribution. Cosine similarity measures the correlation between datasets. Additionally, the Physical Residual is introduced to assess the physical plausibility of the generated data by checking its compliance with the mass conservation principle, which is derived from the Saint-Venant equations. The specific definition of physical residual is given as follows:

The flow of the medium transported in the pipeline necessarily satisfies the laws of conservation of mass, momentum, and energy. Based on the conservation of mass and momentum, and under the assumptions

Table 6

Comparison of generated data quality between PhyCA-GAN and baseline models.

Index	Labels	GAN	WGAN-GP	Phy-GAN	GPSC-GAN	IVQ-LDM	CSGAN-GP	Diff-MTS	PI-CGAN	PCHT-GAN	PhyCA-GAN
MMD	0	0.65	0.39	0.30	0.22	0.26	0.21	0.17	0.14	0.12	0.09
	1	0.73	0.46	0.26	0.34	0.22	0.25	0.13	0.15	0.18	0.15
	2	0.69	0.57	0.38	0.41	0.35	0.18	0.20	0.11	0.15	0.12
	3	0.90	0.54	0.43	0.37	0.31	0.27	0.25	0.23	0.18	0.19
KLD	0	0.058	0.051	0.044	0.046	0.032	0.040	0.028	0.025	0.026	0.021
	1	0.091	0.079	0.067	0.058	0.064	0.043	0.045	0.037	0.025	0.033
	2	0.077	0.063	0.055	0.047	0.044	0.025	0.032	0.020	0.017	0.018
	3	0.064	0.058	0.054	0.041	0.030	0.038	0.030	0.024	0.026	0.022
Cosine	0	0.741	0.805	0.821	0.867	0.880	0.904	0.923	0.919	0.933	0.941
	1	0.767	0.812	0.837	0.843	0.896	0.909	0.915	0.926	0.920	0.934
	2	0.749	0.801	0.825	0.868	0.885	0.913	0.931	0.924	0.921	0.925
	3	0.751	0.818	0.830	0.856	0.875	0.900	0.920	0.920	0.929	0.937
Res	0	0.454	0.341	0.158	0.160	0.055	0.091	0.037	0.034	0.037	0.036
	1	0.517	0.289	0.122	0.154	0.062	0.114	0.060	0.047	0.046	0.043
	2	0.480	0.322	0.167	0.207	0.088	0.143	0.053	0.050	0.046	0.045
	3	0.625	0.393	0.181	0.210	0.093	0.139	0.072	0.067	0.054	0.057

of isothermal conditions and one-dimensional gradually varied flow, the governing equations for unsteady pipe flow are formulated as follows [34]:

$$\begin{cases} \frac{\partial P}{\partial t} + v \frac{\partial P}{\partial x} + \rho a^2 \frac{\partial v}{\partial x} = 0 \\ \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{f|v|}{2D} = 0 \end{cases} \quad (42)$$

where, P is pressure, v is flow velocity, ρ is fluid density, a is the propagation speed of the signal wave, g is gravitational acceleration. f is the Darcy–Weisbach friction factor, D is the inner diameter of the tube.

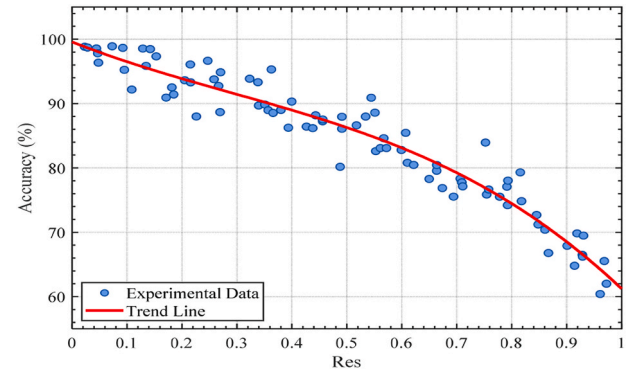
When a leakage occurs, a mass loss is introduced, modifying the continuity equation with an orifice flow mass sink term $L(t)$. To map these equations into the neural network's loss, we define the Res as the violation of this modified continuity equation. The physical residual to be minimized is formulated as:

$$\begin{aligned} \text{Res} &= \frac{\partial X_{\text{gen}}}{\partial t} + \hat{v} \frac{\partial X_{\text{gen}}}{\partial x} + \rho a^2 \frac{\partial v}{\partial x} - L(t) \\ L(t) &= C \sqrt{\frac{2\Delta p(t)}{\rho}} \end{aligned} \quad (43)$$

where, $\Delta p(t)$ is the pressure difference across the leak. C is a constant, which is related to the shape of the orifice and the fluid properties. X_{gen} is the generated data.

It should be noted that evaluating the Res directly from the generated 1D discrete time-series presents the same spatial-gradient challenge. Consistent with the numerical implementation described in section 2.2 (the method for calculating residual term $R(\cdot)$), we use discrete finite difference methods and the traveling wave substitution method to compute the time derivative and the spatial derivative.

Table 6 gives a comprehensive comparison between PhyCA-GAN and baseline models in terms of the quality of generated data. The experimental results demonstrate that PhyCA-GAN achieves the best performance under four different pipeline operation conditions, which is obviously better than all the baseline methods. Notably, it is most advantageous in small leakage scenarios, where improvements over the Diff-MTS model are more than 18% on all metrics. This superior performance can largely be attributed to the synergistic interaction between the proposed KPE module and the multi-scale generation path in PhyCA-GAN: the former can effectively incorporate physical prior knowledge associated with the pipeline operation into the generation process, while the latter can accurately capture multi-level statistical distribution characteristics within the data. While PI-CGAN and PCHT-GAN rank second only to our PhyCA-GAN, their soft physical constraints still encourage rote memorization of data distributions rather than true physical reasoning, resulting in a persistent gap with real data.

**Fig. 6.** Regression curve between Res and accuracy.**Table 7**

Details of the dataset.

	Normal	Large Leak	Medium Leak	Small Leak
Datasets A	2000	400	400	400
Datasets B	2000	200	200	200
Datasets C	2000	100	100	100
Datasets D	2000	50	50	50
Datasets E	2000	20	20	20

The results comprehensively confirm the effectiveness and superiority of PhyCA-GAN in generating sparse leakage samples, which is a new solution for industrial time series data augmentation that combines physical mechanisms and data-driven methods.

We further investigated the direct impact of physical consistency on downstream diagnostic performance. As shown in Fig. 6, there is a clear negative correlation between the physical residual of the generated data and diagnostic accuracy. Samples with lower physical residuals—indicating stricter adherence to physical laws—yield higher and more stable diagnostic accuracies (exceeding 90%). Conversely, as physical errors accumulate, accuracy sharply declines. This trend confirms that the model effectively internalizes physical constraints, providing a robust foundation for downstream tasks.

3.3. Pipeline leak detection

To systematically assess the performance of PhyCA-GAN in imbalanced data conditions, four data sets with different sample ratios were prepared from pipeline data obtained from the experimental platform. The specific configurations of the data sets are summarized in Table 7. Each dataset was randomly divided into a training dataset (70%) and a

Table 8
Average classification results $\pm$ standard deviations (%) across different models.

		GAN	WGAN-GP	Phy-GAN	GPSC-GAN	IVQ-LDM	CSGAN-GP	Diff-MTS	PI-CGAN	PCHT-GAN	PhyCA-GAN
1:5	Acc	61.26 $\pm$ 5.53	72.61 $\pm$ 3.90	79.34 $\pm$ 2.11	88.73 $\pm$ 3.84	92.05 $\pm$ 2.83	92.21 $\pm$ 3.27	97.15 $\pm$ 1.01	97.44 $\pm$ 1.28	97.71 $\pm$ 0.73	97.52 $\pm$ 0.78
	Re	61.49 $\pm$ 2.27	73.44 $\pm$ 1.61	79.89 $\pm$ 1.84	90.05 $\pm$ 5.22	89.38 $\pm$ 4.05	92.50 $\pm$ 1.79	95.29 $\pm$ 0.88	96.17 $\pm$ 1.60	97.85 $\pm$ 2.05	98.08 $\pm$ 0.32
	F1	57.61 $\pm$ 1.35	70.20 $\pm$ 5.54	80.16 $\pm$ 1.27	88.42 $\pm$ 3.49	91.99 $\pm$ 2.54	91.78 $\pm$ 2.64	97.30 $\pm$ 2.12	97.33 $\pm$ 0.87	97.03 $\pm$ 1.10	97.16 $\pm$ 1.77
1:10	Acc	56.15 $\pm$ 3.60	70.35 $\pm$ 4.18	79.57 $\pm$ 1.01	89.36 $\pm$ 2.42	90.63 $\pm$ 1.16	92.45 $\pm$ 1.47	96.43 $\pm$ 0.75	97.29 $\pm$ 0.41	97.05 $\pm$ 0.65	97.17 $\pm$ 0.53
	Re	55.37 $\pm$ 3.29	70.61 $\pm$ 2.21	78.50 $\pm$ 3.37	88.30 $\pm$ 1.59	89.11 $\pm$ 3.37	90.68 $\pm$ 2.52	95.11 $\pm$ 1.29	96.36 $\pm$ 1.04	97.24 $\pm$ 1.12	97.65 $\pm$ 1.07
	F1	53.44 $\pm$ 8.82	69.04 $\pm$ 3.77	78.28 $\pm$ 2.33	88.75 $\pm$ 1.62	89.89 $\pm$ 1.60	91.22 $\pm$ 2.85	95.60 $\pm$ 1.52	96.73 $\pm$ 0.82	97.26 $\pm$ 1.23	97.18 $\pm$ 0.69
1:20	Acc	47.98 $\pm$ 3.38	66.35 $\pm$ 3.99	73.01 $\pm$ 1.81	80.14 $\pm$ 3.15	85.80 $\pm$ 2.35	84.56 $\pm$ 4.41	93.19 $\pm$ 1.86	93.50 $\pm$ 1.77	96.41 $\pm$ 1.04	95.85 $\pm$ 2.02
	Re	43.61 $\pm$ 6.11	66.84 $\pm$ 2.14	72.99 $\pm$ 2.15	78.37 $\pm$ 4.08	85.44 $\pm$ 3.61	86.49 $\pm$ 1.56	92.57 $\pm$ 2.03	93.64 $\pm$ 1.52	93.27 $\pm$ 3.49	93.90 $\pm$ 2.74
	F1	48.29 $\pm$ 2.46	64.70 $\pm$ 5.21	72.53 $\pm$ 2.40	81.42 $\pm$ 1.61	84.71 $\pm$ 3.47	85.71 $\pm$ 2.90	92.25 $\pm$ 1.17	94.90 $\pm$ 2.02	94.55 $\pm$ 1.08	95.11 $\pm$ 1.49
1:40	Acc	43.35 $\pm$ 2.99	58.18 $\pm$ 3.92	65.58 $\pm$ 0.88	71.27 $\pm$ 1.53	77.46 $\pm$ 3.13	79.54 $\pm$ 2.55	87.85 $\pm$ 1.16	91.03 $\pm$ 1.54	90.87 $\pm$ 2.25	91.17 $\pm$ 1.02
	Re	41.61 $\pm$ 5.50	56.35 $\pm$ 7.01	63.92 $\pm$ 1.23	69.41 $\pm$ 4.28	78.30 $\pm$ 1.28	77.19 $\pm$ 3.31	86.37 $\pm$ 1.45	88.88 $\pm$ 4.12	90.12 $\pm$ 1.83	90.72 $\pm$ 1.37
	F1	41.59 $\pm$ 2.26	57.72 $\pm$ 2.38	64.76 $\pm$ 3.33	69.78 $\pm$ 3.06	74.09 $\pm$ 6.76	76.86 $\pm$ 3.26	87.04 $\pm$ 0.73	90.25 $\pm$ 2.02	89.86 $\pm$ 2.44	90.40 $\pm$ 0.44
1:100	Acc	26.81 $\pm$ 2.02	39.59 $\pm$ 4.22	46.34 $\pm$ 2.77	54.10 $\pm$ 3.48	63.19 $\pm$ 1.34	62.77 $\pm$ 1.27	75.72 $\pm$ 3.33	82.09 $\pm$ 0.46	80.80 $\pm$ 2.71	82.27 $\pm$ 1.02
	Re	27.06 $\pm$ 2.66	41.44 $\pm$ 1.78	45.97 $\pm$ 4.19	57.75 $\pm$ 1.96	59.87 $\pm$ 3.28	62.15 $\pm$ 1.02	77.29 $\pm$ 1.57	77.70 $\pm$ 2.82	78.59 $\pm$ 3.30	78.84 $\pm$ 2.31
	F1	25.43 $\pm$ 3.89	41.80 $\pm$ 2.51	47.50 $\pm$ 3.14	55.06 $\pm$ 2.04	60.60 $\pm$ 1.23	61.38 $\pm$ 2.90	75.36 $\pm$ 2.46	76.97 $\pm$ 3.01	77.72 $\pm$ 2.74	79.35 $\pm$ 2.18

test dataset (30%) in order to ensure the statistical validity of the model evaluation. In order to make a fair comparison for all experiments, the model of multi-scale attention CNN (MSACNN) [34] was used as the unified classifier to detect pipeline leaks. The final performance metrics were computed as the average of ten independent experimental runs in order to ensure statistical reliability.

Table 8 summarizes the performance of each model for the downstream fault classification task. The results indicate that under the extreme imbalance condition of 1:40, the accuracy, recall and F1-score of PhyCA-GAN are 91.17%, 90.72% and 90.40%, respectively, which are 4.12%, 4.35% and 2.96% higher than the Diff-MTS model. Furthermore, as the sample ratio is reduced from the ratio of 1:5 to 1:40, PhyCA-GAN shows negligible degradation in performance with an F1-score decrease of less than 8%. Furthermore, all performance metrics outperform the physics-informed baseline models (PI-CGAN and PCHT-GAN). This clearly validates its great robustness to data imbalance. Furthermore, under the extreme 1:100 imbalance ratio, the performance of all models noticeably deteriorates. However, PhyCA-GAN demonstrates the strongest resilience among the evaluated methods, sustaining all metrics above 78%. These results show that by combining physical prior knowledge with data-driven learning, PhyCA-GAN is able to produce samples that are physically consistent and distribution authentic, even under limited sample availability. This approach provides a good solution for enhancing the accuracy and generalization ability of pipeline leak detection models.

3.4. Ablation study

3.4.1. Module ablation

To systematically assess the contribution of each core module in PhyCA-GAN, we designed four configurations of ablation studies that are based on the complete model, as detailed below:

- (1) Model A: Eliminates the KPE module, while retaining all other modules.
- (2) Model B: Keeps only the stochastic turbulence branch from the PDCG module, and the rest of the architecture is the same as the full model.
- (3) Model C: Keeps only the deterministic kinematic branch in the PDCG module and deletes the Stochastic turbulence branch, other modules remain unchanged.
- (4) Model D: Eliminates the physical consistency discrimination branch in the PID module, retaining only the distribution consistency discrimination branch, and keeping the original configuration of all the other components.

Fig. 7 shows the evaluation results of the quality of generated data in dependence on different pipeline operating conditions for the different ablation models. The analysis shows that the KPE module is effective to balance the performance in both physical consistency and statistical distribution fidelity by adaptively integrating physical prior knowledge with data-driven features. Compared to the full PhyCA-GAN model, Model A, which excludes the KPE, has a significant increase in both Res and MMD metrics; The elimination of the deterministic kinematic branch (Model B) results in a notable increase in Res, which is indicative of the degradation of physical plausibility as a result of the absence of physical constraints during the generation process. Conversely, the absence of the stochastic turbulence branch (Model C) results in a reduced diversity and structural content of the resulting data as indicated by severe degradation in the KLD and COSINE measures; In addition, the Model D, where the physical discriminator branch is eliminated, also demonstrates the evident rise of Res, which points to the significance of the given module in the enforcement of physical restrictions on the produced outputs.

As shown in Fig. 8, t-SNE visualization was further used to verify the conclusions obtained from the quantitative analysis. The results show

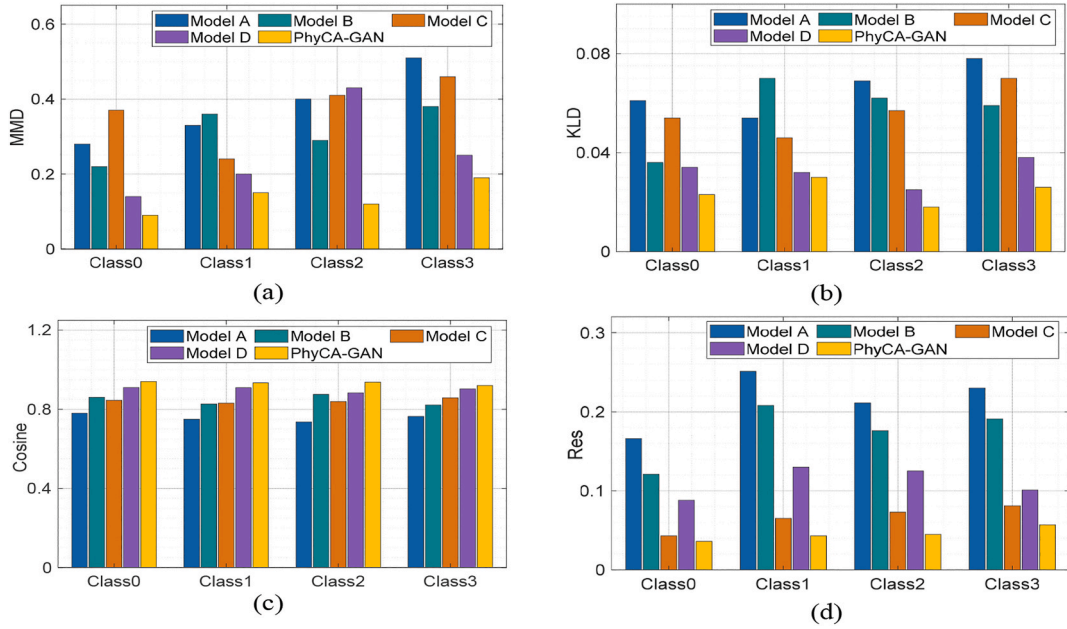


Fig. 7. Quality assessment results of the generated data. (a) MMD. (b) KLD. (c) Cosine. (d) Res.

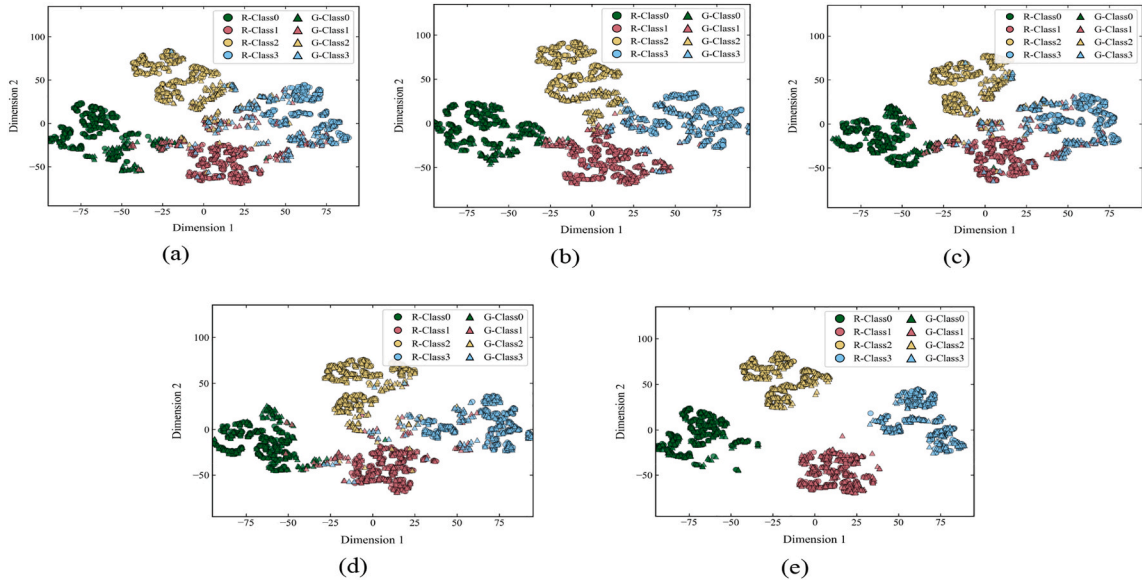


Fig. 8. t-SNE visualization of data structure. (a) Model A. (b) Model B. (c) Model C. (d) Model D. (e) PhyCA-GAN.

that the real samples (labeled as R-Class) form a very consistent and compact structure of clusters. In contrast, the samples produced by each ablation model (labeled as G-Class) show a low overlap with the real samples and are more scattered in the embedded space. For example, the clusters generated by Model A are well separated from the clusters of the real samples, with ambiguous inter-class boundaries. By comparison, the samples generated by PhyCA-GAN exhibit good overlap and compact clustering with the real samples in the embedding space, while maintaining clear separation between different categories. These results can intuitively show the better distribution consistency that PhyCA-GAN has achieved, which can also support the quantitative results that have been discussed above.

3.4.2. Internal component ablation

Further ablation studies were conducted to evaluate the individual contributions of the model's core components:

- (1) Model E: Replaces the dynamic gradient fusion in the KPE module with a fixed-ratio fusion.
- (2) Model F: Omits the mean correction term from the U-Net reverse denoising process in the deterministic kinematic branch.
- (3) Model G: Simplifies the physics-causal attention mechanism in the physics-causal attention fusion module to standard cross-attention.
- (4) Model H: Removes the dynamic gradient coordination mechanism from the generator loss, fixing the adversarial loss weights to a constant.

The performance comparison results for the various ablation models are shown in Fig. 9. When the dynamic gating network is removed (Model E), statistical fidelity drops, and the physical residual increases to over 0.14. This confirms the network's role in ensuring adaptive robustness during feature fusion. Model F experiences a severe

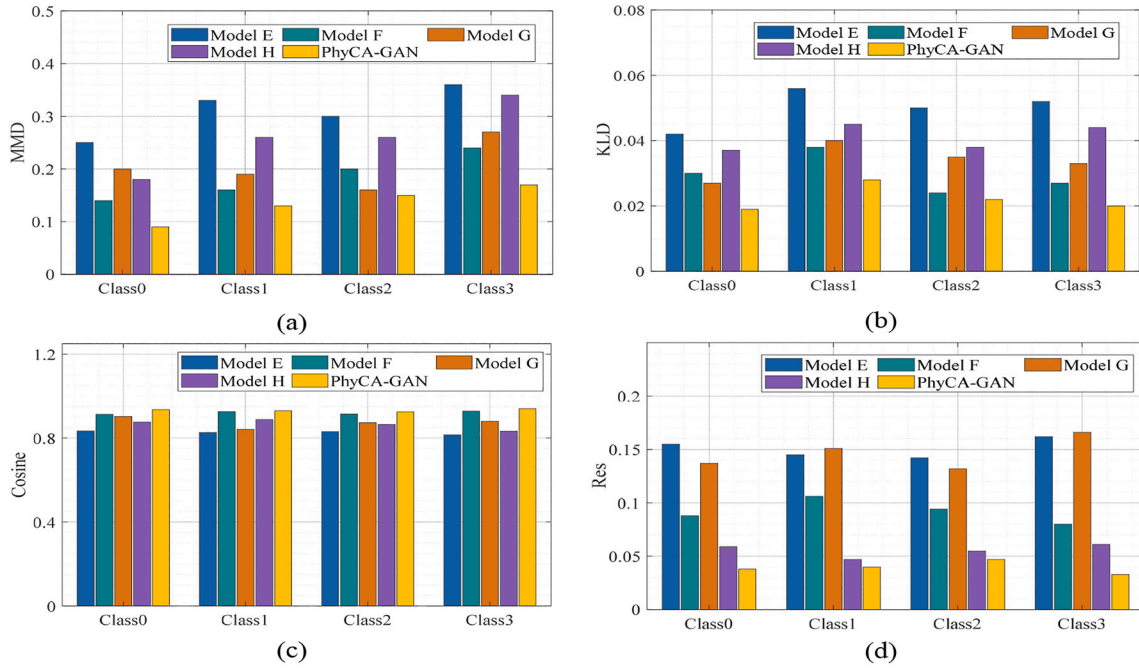


Fig. 9. Quality assessment results of the generated data. (a) MMD. (b) KLD. (c) Cosine. (d) Res.

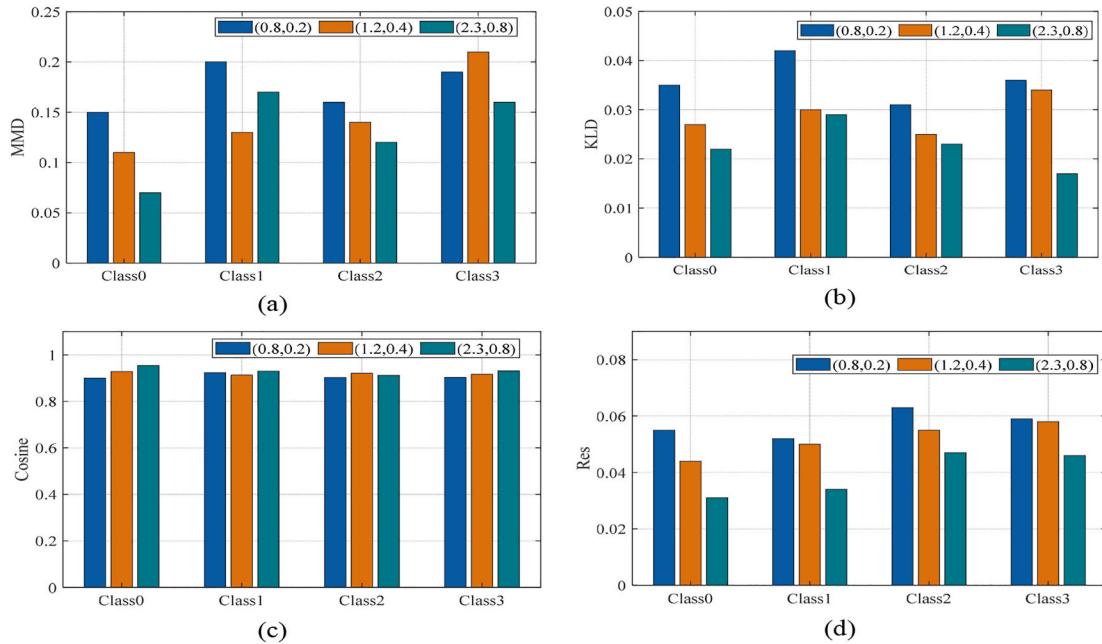


Fig. 10. Quality assessment results of the generated data. (a) MMD. (b) KLD. (c) Cosine. (d) Res.

deterioration in Res despite only a minor drop in statistical fidelity. This suggests that soft loss constraints alone are insufficient to guarantee physical validity during complex diffusion denoising. Furthermore, simplifying the attention mechanism (Model G) leads to a substantial increase in Res compared to the baseline. This demonstrates that physical causal attention acts not merely as a feature fusion tool, but as a hard constraint enforcing the Saint-Venant equations. In Model H, the absence of dynamic gradient coordination causes a marked deterioration in distribution consistency metrics (such as MMD and KLD), underscoring its importance in mitigating inherent gradient conflicts. Ultimately, the proposed PhyCA-GAN achieves the best performance across all metrics, validating the necessary synergy of these components.

3.5. Analysis of generalizability and robustness

3.5.1. Analysis under varying operating conditions

We evaluated the generalizability of PhyCA-GAN across three pipeline pressure conditions, with the resulting data quality metrics presented in Fig. 10. Across all pressure combinations, the MMD values remain consistently low and the KLD values stay below 0.042, confirming that the generated data closely matches the true distribution. Furthermore, the Cosine similarities exceed 0.9, and the physical residuals (Res) stay stable under 0.079. These findings confirm the model's ability to accurately reconstruct fluid dynamics features under varying pressures, highlighting its robust shape reproduction and operational

Table 9Average classification results $\pm$ standard deviations (%) across different pressure.

Pressures/MPa	Index	1:5	1:10	1:20	1:40	1:100
(0.8, 0.2)	Acc	96.11 $\pm$ 1.73	95.50 $\pm$ 0.61	93.72 $\pm$ 2.29	89.45 $\pm$ 0.71	79.28 $\pm$ 1.53
	Re	95.79 $\pm$ 2.02	96.72 $\pm$ 1.38	92.99 $\pm$ 1.47	88.78 $\pm$ 1.12	81.00 $\pm$ 0.28
	F1	96.63 $\pm$ 0.77	95.05 $\pm$ 0.86	93.24 $\pm$ 0.69	87.50 $\pm$ 1.68	79.66 $\pm$ 1.81
(1.2, 0.5)	Acc	97.17 $\pm$ 0.32	96.55 $\pm$ 2.01	95.07 $\pm$ 0.25	90.90 $\pm$ 0.73	80.51 $\pm$ 3.17
	Re	97.48 $\pm$ 0.75	96.83 $\pm$ 0.99	93.46 $\pm$ 1.12	91.54 $\pm$ 3.00	79.70 $\pm$ 1.64
	F1	96.63 $\pm$ 1.40	95.95 $\pm$ 1.06	94.72 $\pm$ 1.84	88.38 $\pm$ 2.94	80.09 $\pm$ 1.23
(2.3, 0.8)	Acc	97.60 $\pm$ 0.88	98.25 $\pm$ 0.25	94.31 $\pm$ 3.62	92.09 $\pm$ 1.37	81.60 $\pm$ 2.08
	Re	98.15 $\pm$ 1.31	97.01 $\pm$ 1.11	94.96 $\pm$ 1.75	89.52 $\pm$ 3.72	82.22 $\pm$ 0.53
	F1	97.74 $\pm$ 0.39	96.49 $\pm$ 2.53	93.28 $\pm$ 1.90	90.64 $\pm$ 1.36	80.49 $\pm$ 1.37

Table 10

Model performance analysis under different noise conditions.

SNR (dB)	MMD	KLD	Cosine	Res	Acc (%)	F1 (%)
10	0.08	0.017	0.949	0.030	97.18 $\pm$ 1.44	98.42 $\pm$ 1.37
5	0.12	0.018	0.942	0.036	96.30 $\pm$ 2.52	97.47 $\pm$ 2.12
0	0.15	0.024	0.936	0.047	96.65 $\pm$ 2.08	97.51 $\pm$ 1.70
-5	0.21	0.045	0.925	0.043	94.07 $\pm$ 0.71	93.69 $\pm$ 3.54
-10	0.33	0.047	0.901	0.055	91.91 $\pm$ 1.38	91.77 $\pm$ 2.99

Table 11

Average performance metrics under different wave velocity estimation errors.

	+20%	+10%	0%	-10%	-20%
MMD	0.22	0.17	0.11	0.15	0.25
KLD	0.045	0.029	0.020	0.023	0.037
Cosine	0.901	0.924	0.933	0.927	0.910
Res	0.084	0.061	0.045	0.058	0.078

stability.

Table 9 presents the pipeline leak diagnosis performance across various data imbalance ratios. At a 1:5 ratio, the accuracy exceeds 96% across all operating conditions. As the imbalance worsens to 1:20 and 1:40, performance remains robust, with all metrics staying between 87% and 95%. Remarkably, even at an extreme 1:100 imbalance, the accuracy holds steady near 80%. These results confirm that the high-fidelity samples generated by PhyCA-GAN successfully bridge data gaps in rare operating conditions, thereby improving the reliability of downstream diagnostic tasks.

3.5.2. Noise robustness analysis

The performance of PhyCA-GAN in complex noisy environments was evaluated by measuring generation quality and diagnostic accuracy across SNRs from 10 dB to -10 dB. The average results for each metric are shown in Table 10. Although metrics decline slightly as noise increases, overall performance remains highly competitive. Under severe interference (-10 dB), MMD and KLD are recorded at 0.33 and 0.047, respectively, proving that the model successfully isolates the true distribution of fluid dynamics features from background noise. In

downstream leakage detection, accuracy exceeds 96% at SNRs of 0 dB and above. Even under the extreme -10 dB condition, Acc reaches 91.91% alongside an F1 score of 91.77%, confirming the model's exceptional noise robustness and practical adaptability.

3.5.3. Parameter sensitivity analysis

To assess the robustness of the dynamic kinematic mask against wave velocity estimation errors in the physics-causal attention fusion module, we introduced perturbations between -20% and +20% during inference. As summarized in Table 11, while extreme deviations ($\pm 20\%$) slightly increase the physical residual (Res) to a maximum of 0.084, overall statistical fidelity remains high. Specifically, MMD stays below 0.25, Cosine similarity exceeds 0.9, and KLD is kept under 0.045. This stability stems from the soft mapping mechanism used during mask generation, which prevents the abrupt truncation of valid spatio-temporal correlations. Consequently, PhyCA-GAN consistently generates high-quality data even with imprecise wave velocity estimates.

3.6. Industrial validation

In order to verify the robustness and generalization ability of PhyCA-GAN in industrial scenarios, leakage diagnosis experiment was carried out on an operating transmission pipeline operated by an oil transportation company. The pipeline segment that was used in the experiment covers a distance of 10.2 km with a nominal diameter of DN300. The experimental platform employs a controllable release mechanism to simulate leaks. Each release is kept brief to ensure the safe recovery of the medium. During a leak event, acoustic signals are recorded at 1000Hz by two sensors located at the pipeline's inlet and outlet. Normal samples are collected from operational data recorded over several days before and after the experiment. Fig. 11 illustrates the complete experimental setup. To reduce the effects of randomness, the results presented are the arithmetic mean of 10 independent experimental runs, with detailed results presented in Table 12.

As summarized in Table 12, PhyCA-GAN is able to achieve optimal or near-optimal performance in industrial oil pipeline leak detection for varying sample imbalance ratios. Specifically, in all the imbalance conditions evaluated from 1:5 to 1:40, PhyCA-GAN significantly outperforms all the baseline models in terms of accuracy, recall and F1-

**Fig. 11.** Industrial pipeline experiment platform.

Table 12
Average classification results $\pm$ standard deviations (%) across different models.

		GAN	WGAN-GP	Phy-GAN	GPSC-GAN	IVQ-LDM	CSGAN-GP	Diff-MTS	PI-CGAN	PCHT-GAN	PhyCA-GAN
1:5	Acc	56.71 $\pm$ 7.08	69.83 $\pm$ 5.45	78.50 $\pm$ 3.84	87.52 $\pm$ 4.17	90.77 $\pm$ 3.58	92.08 $\pm$ 4.40	97.13 $\pm$ 1.34	96.55 $\pm$ 2.02	97.18 $\pm$ 0.85	97.35 $\pm$ 1.08
	Re	54.92 $\pm$ 4.61	70.80 $\pm$ 2.17	77.29 $\pm$ 5.10	89.89 $\pm$ 2.01	89.60 $\pm$ 2.16	91.92 $\pm$ 1.52	96.10 $\pm$ 1.58	96.79 $\pm$ 1.35	97.22 $\pm$ 0.23	98.49 $\pm$ 0.75
	F1	56.24 $\pm$ 2.82	70.27 $\pm$ 3.52	79.03 $\pm$ 3.42	89.15 $\pm$ 3.37	88.38 $\pm$ 4.41	91.43 $\pm$ 2.18	96.86 $\pm$ 2.73	97.35 $\pm$ 0.77	98.34 $\pm$ 1.18	97.02 $\pm$ 1.42
1:10	Acc	54.54 $\pm$ 4.30	69.85 $\pm$ 2.69	78.49 $\pm$ 1.73	87.64 $\pm$ 4.05	90.42 $\pm$ 1.79	91.62 $\pm$ 0.83	96.49 $\pm$ 0.25	95.80 $\pm$ 2.23	96.62 $\pm$ 0.73	97.10 $\pm$ 0.83
	Re	52.86 $\pm$ 6.86	70.03 $\pm$ 1.57	77.77 $\pm$ 3.19	88.78 $\pm$ 1.22	88.88 $\pm$ 2.54	89.59 $\pm$ 3.27	94.36 $\pm$ 1.72	96.25 $\pm$ 1.61	95.71 $\pm$ 1.00	96.47 $\pm$ 1.21
	F1	52.63 $\pm$ 4.47	69.26 $\pm$ 2.24	76.35 $\pm$ 3.23	86.19 $\pm$ 3.36	89.35 $\pm$ 1.38	90.51 $\pm$ 2.40	95.70 $\pm$ 2.09	95.71 $\pm$ 2.84	96.90 $\pm$ 1.43	95.62 $\pm$ 1.46
1:20	Acc	46.34 $\pm$ 1.97	66.62 $\pm$ 2.68	73.54 $\pm$ 2.40	80.50 $\pm$ 1.63	82.28 $\pm$ 3.76	84.21 $\pm$ 3.94	92.75 $\pm$ 1.21	92.24 $\pm$ 3.38	93.95 $\pm$ 2.21	94.64 $\pm$ 1.42
	Re	45.88 $\pm$ 4.20	65.66 $\pm$ 2.83	71.88 $\pm$ 3.95	79.73 $\pm$ 3.48	84.15 $\pm$ 0.50	84.72 $\pm$ 1.78	92.02 $\pm$ 1.47	93.12 $\pm$ 1.31	93.72 $\pm$ 1.10	93.78 $\pm$ 1.83
	F1	45.01 $\pm$ 3.51	64.18 $\pm$ 4.05	72.46 $\pm$ 5.57	80.16 $\pm$ 1.25	82.62 $\pm$ 3.78	84.46 $\pm$ 0.77	91.38 $\pm$ 1.05	92.06 $\pm$ 1.16	92.99 $\pm$ 2.34	93.42 $\pm$ 2.22
1:40	Acc	40.73 $\pm$ 3.56	57.40 $\pm$ 2.28	64.87 $\pm$ 4.72	71.64 $\pm$ 0.78	75.97 $\pm$ 1.42	78.04 $\pm$ 3.17	87.47 $\pm$ 1.65	86.87 $\pm$ 3.33	90.60 $\pm$ 2.28	91.33 $\pm$ 0.36
	Re	42.28 $\pm$ 2.11	56.64 $\pm$ 5.33	64.14 $\pm$ 6.17	70.08 $\pm$ 3.50	77.15 $\pm$ 1.01	75.61 $\pm$ 5.83	87.19 $\pm$ 0.81	89.41 $\pm$ 1.68	89.35 $\pm$ 1.90	90.06 $\pm$ 1.51
	F1	41.45 $\pm$ 2.90	56.26 $\pm$ 3.74	65.00 $\pm$ 3.33	69.37 $\pm$ 4.51	75.32 $\pm$ 3.85	76.67 $\pm$ 3.35	87.90 $\pm$ 0.53	90.08 $\pm$ 1.35	90.19 $\pm$ 2.05	89.72 $\pm$ 0.85
1:100	Acc	24.59 $\pm$ 5.05	40.40 $\pm$ 3.84	47.50 $\pm$ 3.47	56.08 $\pm$ 3.38	63.66 $\pm$ 1.62	63.28 $\pm$ 2.50	76.07 $\pm$ 0.84	78.22 $\pm$ 3.06	80.43 $\pm$ 1.31	80.88 $\pm$ 3.26
	Re	25.81 $\pm$ 3.27	39.62 $\pm$ 4.18	48.08 $\pm$ 3.01	56.42 $\pm$ 2.80	61.10 $\pm$ 4.15	62.04 $\pm$ 3.38	74.92 $\pm$ 5.34	77.56 $\pm$ 3.74	77.67 $\pm$ 4.38	82.15 $\pm$ 0.89
	F1	23.63 $\pm$ 3.42	42.08 $\pm$ 1.11	45.27 $\pm$ 4.48	55.95 $\pm$ 4.29	60.17 $\pm$ 3.93	62.11 $\pm$ 4.26	76.65 $\pm$ 3.90	77.19 $\pm$ 4.06	78.49 $\pm$ 2.63	79.02 $\pm$ 2.53

score. Notably, it shows very little performance degradation with increasing severity of imbalance. For example, with the extreme 1:40 imbalance setting, PhyCA-GAN achieves an accuracy of 91.33%, which is about 0.73% better than the second-best model, PCHT-GAN (90.60%). It also has a better recall (90.06%) and F1-score (89.72%). Furthermore, even under extreme imbalance conditions of 1:100, PhyCA-GAN still outperforms other models across all metrics. These results confirm that PhyCA-GAN has a high level of generalization ability and robustness, which proves that it can be used in complex leak detection tasks in real industrial pipeline environments.

3.7. Computational complexity analysis

To evaluate deployment feasibility, we analyzed the computational complexity of the models on a single NVIDIA RTX 4060 GPU with a batch size of 64. Table 13 compares the model parameters, FLOPs, and the inference time required to generate 1000 samples. As expected, single-step feedforward models like GAN, WGAN-GP, and CSPP-GAN show lower computational overhead. Conversely, IVQ-LDM and Diff-MTS require higher FLOPs and longer inference times due to their iterative denoising processes. Owing to the integration of physical governing equations, both PI-CGAN and PCHT-GAN exhibit higher FLOPs and more params compared to conventional GAN-based models. By embedding explicit fluid dynamics boundaries via the KPE and the physical causal attention mechanism, PhyCA-GAN effectively narrows the data manifold search space. This allows the deterministic branch to achieve high physical fidelity with fewer iterative steps, mitigating the typical computational bottleneck of diffusion models.

4. Conclusions

This paper introduces PhyCA-GAN to bridge the gap between data-driven models and physical laws in pipeline leak diagnosis. By building spatiotemporal causality directly into the model, our framework generates high-quality leakage data that follows fluid dynamics. In particular, The KPE acts as a physical guide. Unlike standard encoders that just fuse features, the KPE extracts clear kinematic priors from a learnable knowledge base. This sets physical boundaries early on, ensuring the generated signals follow transmission limits from the start; The PDCG reconciles physical tendencies and accidents; It separates the signal into a deterministic kinematic component, and a stochastic turbulence component. This enables the model to describe the overall propagation patterns, as well as the high-frequency noise. In addition, non-physical artifacts are eliminated by the physics-causal attention mechanism, which prevents impossible correlations; The PID guarantees physical and statistical accuracy of the data. Its decoupled architecture enables the generator to receive independent gradient feedback regarding physical consistency and statistical authenticity. This guides the model to learn the true data distribution while following conservation laws, preventing the generation of "plausible but physically wrong" samples; Extensive experiments on two platforms confirm that PhyCA-GAN data synthesized is significantly superior to existing baseline augmentation systems.

CRedit authorship contribution statement

Yongqiang Zhu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Shuaiyong Li:** Supervision, Funding acquisition, Formal analysis. **Xianming Lang:** Software, Data curation, Conceptualization. **Liang Liu:** Writing – review & editing, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

Table 13

Model computational complexity metrics.

	GAN	WGAN-GP	GPSC-GAN	IVQ-LDM	CSGAN-GP	Diff-MTS	PI-CGAN	PCHT-GAN	PhyCA-GAN
Params (M)	2.1	2.5	4.3	12.5	3.5	15.8	18.3	25.7	11.2
FLOPs (G)	0.8	1.1	2.4	18.7	1.8	22.4	11.6	10.8	15.3
Inference Time (s)	0.2	0.3	0.9	15.2	0.7	18.5	15.1	13.3	13.8

the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Shao X, Cai B, Ahmed S, Zhou X, Hu Z, Sui Z, Liu X. Towards proactive corrosion management: a predictive modeling approach in pipeline industrial applications. *Process Saf Environ* 2024;190:1471–80.
- [2] Lin Z, Wang Y, Fu X, Zou Z, Feng L, Fang L, Qing X. Defect localization and quantification of energy transportation pipeline based on multiscale 1DCNN and BiLSTM with multidimensional attention mechanism. *Energy* 2025;344:139879.
- [3] Wang J, Wang T, Ma D. Pipeformer: a multi-channel patching-based transformer method with pyramid cross-attention for leak detection of energy transportation system. *Energy* 2025;335:138096.
- [4] Cai M, Liu Y, Zhao J, Wang W, Pedrycz W. Physics-informed generative self-attention framework for sliding bearing fault diagnosis. *IEEE Sens J* 2026;26: 5792–804.
- [5] Jiang Y, Guan L, Liu W, Yu Y. Jamming waveform generation method based on generative adversarial network model. *IEEE Signal Process Lett* 2024;31:1114–8.
- [6] Zhu Y, Li S, Lang X, Liu L. PCHT-GAN: physics-guided adaptive fusion with dynamic low-rank attention for pipeline leak diagnosis under imbalanced data. *Inf Fusion* 2025;127:103802.
- [7] Ge L, Sadhu A. Domain adaptation for structural health monitoring via physics-informed and self-attention-enhanced generative adversarial learning. *Mech Syst Signal Process* 2024;211:111236.
- [8] Zheng M, Wang F, Hu X, Hu L, Yu Q, Zheng X. UFIDSF: an undersampling approach based on feature importance and double side filter for imbalanced data classification. *Future Gener Comput Syst* 2025;167:107750.
- [9] Xinmin T, Annan X, Lihang S, Junxuan L, Xinyue G, Sirui T. Density clustering hypersphere-based self-adaptively oversampling algorithm for imbalanced datasets. *Knowl-Based Syst* 2025;329:114407.
- [10] Zhou C, Zou L. Adaptive synthetic oversampling with Poincaré distance and two-stage weight assignment. *Neurocomputing* 2025;647:130436.
- [11] Sun Z, Ying W, Zhang W, Gong S. Undersampling method based on minority class density for imbalanced data. *Expert Syst Appl* 2024;249:123328.
- [12] Zhu Q, Wang X, Zhang N, Xu Y, He Y. Novel k-medoids based SMOTE integrated with locality preserving projections for fault diagnosis. *IEEE Trans Instrum Meas* 2022;71:1–8.
- [13] Tang H, Shao L, Sebe N, Van Gool L. Enhanced multi-scale cross-attention for person image generation. *IEEE T Pattern Anal* 2025;47:3377–93.
- [14] Hu J, Cao Y, Tan G. A dynamic spatiotemporal graph generative adversarial network for scenario generation of renewable energy with nonlinear dependence. *Energy* 2025;335:138049.
- [15] Chen L, Xu Y, Zhu Q, He Y. Adaptive multi-head self-attention based supervised VAE for industrial soft sensing with missing data. *IEEE Trans Autom Sci Eng* 2024; 21:3564–75.
- [16] He X, Liu T, Zhang Y, Xie R. Soft sensor modeling based on vector-quantized weighted-Wasserstein VAE for polyester polymerization process. *IEEE Trans Ind Inf* 2024;20:11338–47.
- [17] Zhang R, Fares A, Tijani IA, Zayed T, Jin Z, Yussif AM. Acoustic based leak location detection for water supply pipelines in urban areas via multi-task deep learning. *Eng Appl Artif Intell* 2026;163:112718.
- [18] Hassani S, Mustapha S, Mousavi M, Li J, Song C, Dackermann U. Enhancing structural condition assessment in steel pipelines via a WGAN-AAE data fusion methodology. *Inf Fusion* 2025;129:103970.
- [19] Li Q, Gao X, Lu H, Li B, Zhai F, Wang T, Hao Y. A multimodal data generation method for imbalanced classification with dual-discriminator constrained diffusion model and adaptive sample selection strategy. *Inf Fusion* 2025;117:102843.
- [20] Zhang Q, Hu Z, Wang Y, Zhang H, Di J. Video and noise collaboratively guided semi-supervised diffusion model for video action detection. *Pattern Recogn* 2026; 169:111925.
- [21] Yang J, Xie J, Gao X, Fu K, Zhu J, Wang J. Synthetic magnetic flux leakage signal generation using diffusion models: a novel approach to improve pipeline defect detection accuracy. *J Pipeline Sci Eng* 2025:100363.
- [22] Xu H, Li Y, Wang D, Jing Q. A spatiotemporal concentration field reconstruction method for natural gas leakage based on the integration of diffusion models and PIGCN. *Reliab Eng Syst Saf* 2026;266:111811.
- [23] Chen X, Fu M, Liu X, Zhu J. Synthesizing labeled magnetic flux leakage signals for pipeline integrity assessment: a generation and evaluation methodology. *IEEE Trans Instrum Meas* 2025;74:1–17.
- [24] Yin J, Wang Y, Xu Z, Yang Y, Zhu M. MaSGAN: a malfunction sound detection model for industrial machines using the generative adversarial networks. *Expert Syst Appl* 2025;293:128602.
- [25] Hu X, Zhang H, Ma D, Wang R. A tnGAN-based leak detection method for pipeline network considering incomplete sensor data. *IEEE Trans Instrum Meas* 2021;70: 1–10.
- [26] Zhang J, Cui X, Dai S, Li J, Li L. DCDGNet: dual-frequency cross-feature diffusion GAN for single fusion sonar image generation in exposed subsea pipeline inspection. *Mar Pollut Bull* 2026;225:119289.
- [27] Wang C, Wang Z, Ma L, Dong H, Sheng W. Subdomain-alignment data augmentation for pipeline fault diagnosis: an adversarial self-attention network. *IEEE Trans Ind Inf* 2024;20:1374–84.
- [28] Liu B, Ding L, He L, Yang L, Zhang X, Tian Y. Residual generative adversarial network-driven data enhancement for magnetic flux leakage-based defect recognition in oil and gas pipelines. *Eng Appl Artif Intell* 2025;162:112720.
- [29] Yang W, Zhang H, Lim JB, Zhang Y, Meng H. A new chiller fault diagnosis method under the imbalanced data environment via combining an improved generative adversarial network with an enhanced deep extreme learning machine. *Eng Appl Artif Intell* 2024;137:109218.
- [30] Lyu P, Cheng Y, Zhang M, Yu W, Xia L, Liu C. GPSC-GAN: a data enhanced model for intelligent fault diagnosis. *IEEE Trans Instrum Meas* 2024;73:1–16.
- [31] Zhang T, Lin J, Jiao J, Zhang H, Li H. An interpretable latent denoising diffusion probabilistic model for fault diagnosis under limited data. *IEEE Trans Ind Inf* 2024; 20:10354–65.
- [32] Zhang S, Xu J, Wang W, Gong Z. Application of an improved GAN with a filter mechanism in small-sample bearing fault diagnosis. *Measurement* 2026;257: 118583.
- [33] Ren L, Wang H, Laili Y. Diff-MTS: temporal-augmented conditional diffusion-based AIGC for industrial time series toward the large model era. *IEEE Trans Cybern* 2024;54:7187–97.
- [34] Zhu Y, Li S, Lang X, Liu L. Physics-informed CGAN and multi-scale attention CNN for pipeline leakage diagnosis under imbalanced data. *Adv Eng Inform* 2025;66: 103471.
- [35] Zhu Y, Li S, Lang X, Liu L. PCHT-GAN: physics-Guided adaptive fusion with dynamic low-rank attention for pipeline leak diagnosis under imbalanced data. *Inf Fusion* 2025;127:103802.